Rebotic — Hardware Parts List

Conversational expo robot · laptop = mind · body moves: jaw, neck, head, two arms, LED eyes · wired USB audio

Prepared 2026-07-20 · matches body/specs/03-electronics.md · prices are approximate USD — local market prices vary

1 · Controller (the body's motion brain)

Part (market name)	Qty	Description	Purpose in the robot	Approx.
ESP32 DevKit V1 (30-pin) *search: "ESP32 DevKit V1 WROOM-32 development board"*	1	Small WiFi-enabled microcontroller board with USB port. Any ESP32-WROOM-32 dev board works.	Sits inside the body. Receives mood commands from the laptop over WiFi and drives all servos, eyes, and the button.	$5–8

2 · Movement (jaw, neck, head, arms)

Part (market name)	Qty	Description	Purpose in the robot	Approx.
SG90 9g micro servo *search: "Tower Pro SG90 9g micro servo motor"*	5 + 2 spare	Tiny hobby servo, ~180° rotation, plastic gears. Comes with horns (arms) and mounting screws.	One each for: jaw (talks), neck pan (turns left/right), head tilt (nods), left arm, right arm (shrug). Spares are mandatory — plastic gears strip on long demo days.	$2–3 each
Pan/tilt servo bracket kit (SG90 size) *search: "SG90 pan tilt bracket kit 2 axis"*	1	Two-axis plastic mounting frame that holds two SG90s stacked.	The neck mechanism — bottom servo turns the head, top servo nods it.	$3–5
Pushrod / stiff steel wire, ~1mm *search: "RC pushrod linkage 1mm" (or use a straightened paper clip)*	1 pack	Stiff wire link between a servo horn and a moving part.	Connects the jaw servo to the hinged lower jaw — the mouth linkage.	$1–3

3 · Eyes & input

Part (market name)	Qty	Description	Purpose in the robot	Approx.
WS2812B 8-LED ring (5V) *search: "WS2812B 8 bit LED ring 5V NeoPixel"*	2	Ring of 8 individually-addressable RGB LEDs, single data wire, chainable.	The eyes — blink randomly, dim cyan at rest, bright white when attending, pulsing amber	$2–4 each

			while thinking, eye-roll on shrug.	
Arcade push button, 60mm, momentary *search: "60mm arcade button momentary LED" (normally-open)*	1	Big springy button, the kind on arcade machines. Momentary / normally-open type.	Push-to-talk: visitors hold it while speaking. Mounted on the desk/stand — big and satisfying, because people will mash it.	$3–6

4 · Ears & voice (USB audio — plugs into the LAPTOP, never the ESP32)

Part (market name)	Qty	Description	Purpose in the robot	Approx.
USB directional microphone *search: "USB cardioid gooseneck microphone" or "mini USB conference mic cardioid"*	1	Compact plug-and-play USB mic. **Cardioid/directional pattern is the priority** — it hears what's in front, rejects room noise. Worth spending extra here.	The robot's ears — mounted in the head/chest facing the visitor. The laptop listens through it via a USB cable.	$10–25
USB-powered mini speaker, 3–5W *search: "USB powered mini speaker 5V wired" (USB sound, or 3.5mm audio + USB power)*	1	Small full-range wired speaker powered over USB. 3–5W is plenty for one visitor at arm's length.	The robot's voice — mounted behind a grille in the chest, so sound and jaw movement come from the same body.	$8–15
USB 2.0 extension cable, 3m (A-male to A-female) *search: "USB 2.0 extension cable 3m"*	2	Plain USB extension leads.	The hidden "tubes" — route mic and speaker from the body down the stand to the laptop. Plug into two separate laptop ports (not one hub).	$3–5 each

5 · Power

Part (market name)	Qty	Description	Purpose in the robot	Approx.
5V 5A DC power adapter (mains) *search: "5V 5A power supply adapter 5.5mm barrel jack"*	1	Wall adapter, 5 volts at 5 amps, barrel plug.	Dedicated servo + LED power rail. Five stalled servos pull ~3.3A — **never** power servos from the ESP32 board, it will brown-out and reboot mid-demo.	$8–12

DC barrel jack to screw-terminal adapter (female) *search: "DC 5.5x2.1 female barrel jack screw terminal adapter"*	1	Lets you connect bare wires to the adapter's barrel plug, no soldering.	Taps the 5V supply into the servo power distribution wiring.	$1–2
Electrolytic capacitor, 1000µF / 10V *search: "1000uF 10V electrolytic capacitor"*	1 (buy 2–3)	Large can-shaped capacitor. Mind the polarity stripe when wiring.	Sits across the servo power rail — absorbs the current spike when servos start moving so nothing resets.	<$1
Resistor, 470Ω, 1/4W *search: "470 ohm resistor 1/4W" (usually sold in packs)*	1 pack	Ordinary small resistor.	Goes in series with the eyes' LED data line — protects the first pixel from signal spikes.	<$1
Micro-USB data cable *search: "micro USB data cable" (must be DATA cable, not charge-only)*	1	Standard phone-style cable — verify it carries data.	Powers the ESP32's logic from the laptop and is how firmware gets flashed onto it.	$2–3

6 · Assembly & body shell

Part (market name)	Qty	Description	Purpose in the robot	Approx.
Dupont jumper wire set (M-M, M-F, F-F) *search: "dupont jumper wires 120pcs kit"*	1 kit	Pre-crimped hookup wires for breadboard/header connections.	All wiring between ESP32, servos, LEDs, and button. Buy the big kit — spares are on the expo ship list.	$4–6
Breadboard, half-size (400 tie points) *search: "solderless breadboard 400 points"*	1	Solderless prototyping board.	Holds the ESP32 and the power distribution — no soldering needed for the whole build.	$3–5
Foam board / foam-core sheets, 5mm *search: "foam board A3 5mm" (art/stationery shop)*	2–3 sheets	Lightweight rigid board, cuts with a knife.	The head shell and both arms. **Must be light** — SG90 servos can't swing heavy parts, and every gram costs servo lifetime.	$3–6
Acoustic / packing foam scrap *search: "acoustic foam panel small" (or reuse packaging foam)*	small piece	Soft open-cell foam.	Isolation pad between speaker and mic mounts — stops the robot hearing its own voice and interrupting itself.	$0–3

Hot glue gun + sticks, zip ties, electrical tape *(household / any hardware shop)*	—	General assembly supplies.	Mounting servos in foam, strain-relieving the cables where they exit the body — tugged cables must not end the demo.	$5–8

> **Estimated total: ~$55–90** · The USB microphone is the one place to spend more — a genuinely directional mic is the biggest single upgrade to demo quality in a noisy hall.

Buying tips

- **Already owned, not on the list:** the laptop (the mind), a phone (WiFi hotspot), and a wired headset mic (expo fallback input — keep it in the bag).
- **Servos:** buy all 7 from the same seller/batch so they behave identically. "MG90S" (metal gear) is a worthy ~$1/pc upgrade for the two arm servos if available.
- **Speaker:** if it has a 3.5mm audio plug + USB power plug instead of pure USB audio, that's fine — the laptop has both sockets.
- **Power adapter:** don't buy 5V 3A to save two dollars — 5A is the floor for five servos.
- **Capacitor polarity:** electrolytic caps have a marked negative stripe — wiring one backwards destroys it.
- **Micro-USB cable:** test it transfers data (connect a phone) — charge-only cables are the classic "board won't flash" trap.
